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RESEARCH PAPER

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The biter bit; scavenging behaviour of native freshwater crayfish on the carrion of native and introduced fish predators in Aotearoa-New Zealand

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Abstract – Freshwater crayfish are important prey items for many fish but are themselves opportunistic scavengers of carcasses of predatory fish. In Aotearoa-New Zealand, kōura, the native freshwater crayfish *Paranephrops planifrons*, a culturally treasured 'taonga' species for indigenous Māori, suffer heavy predation from introduced fish. We investigated potential differences in kōura scavenging behaviour towards carrion of a co-evolved native predator, the long-finned eel *Anguilla dieffenbachia* and a non-native catfish predator, the brown bullhead *Ameiurus nebulosus*. Laboratory experiments showed kōura readily leave refugia to feed on dead predatory fish, with little discrimination between eel, catfish or marine (control) fish carcasses. There was no difference in kōura consumption rate of different fish and no difference in the calorific assimilation rate between eel and catfish. Our results suggest kōura will scavenge the carcasses of catfish that have died within invaded systems, even if this trophic interaction is not acknowledged in freshwater food web studies. Local catfish eradication programmes result in culls of thousands of catfish annually. These carcasses are currently wasted and we suggest they could be repurposed as a food resource for kōura, particularly in a future conservation-based aquaculture setting. This could help offset negative impacts the invader is currently having on native crayfish populations.

Keywords: Aotearoa-New Zealand / catfish / invasive fish / kōura / scavenging behaviour

1 Introduction

Freshwater crayfish can be major components of predatory fish diets (Stein and Magnuson, 1976; Barnes and Hicks, 2003; Reynolds, 2011). However, crayfish also consume fish themselves, whether functioning as a predator of live fish, fish eggs or a scavenger of fish carcasses (Guan and Wiles, 1998; Parkyn *et al.*, 2001; Burks and Lodge, 2002; Taylor and Soucek, 2010; Reynolds, 2011; Unger and Hickman, 2019; Boros *et al.*, 2020). Freshwater crayfish are also indicator species for general freshwater ecosystem health and a keystone species for ecosystem processes (Dorn and Wojdak, 2004; Thomas and Taylor, 2013; Parkyn and Kusabs, 2014; Boros *et al.*, 2020). In the functional role of scavenger, crayfish feeding on fish carcasses can regulate nutrient dynamics in freshwater ecosystems (Boros *et al.*, 2020). The mechanisms involved include release of nutrients from carcasses to higher trophic levels via incorporation into crayfish tissue and excretion, as well as bioturbation and feeding activities impacting nutrient cycling (Lipták *et al.*, 2019; Boros *et al.*, 2020).

Aotearoa-New Zealand (hereafter Aotearoa-NZ), has two species of native freshwater crayfish, both referred to by the same name, kōura (traditional knowledge of Māori the indigenous people of Aotearoa-New Zealand, not distinguishing between the two species – McDowall, 2011). These are the northern kōura, *Paranephrops planifrons*, which occurs in the North Island and the West Coast of the South Island and the southern kōura, *Paranephrops zealandicus,* which occurs in the east and south of the South Island. Kōura are a taonga 'culturally treasured' species for Māori, prized as a food resource and for iconic and heritage values (Hiroa, 1921; Kusabs and Quinn, 2009).

Over the past two centuries, many non-native freshwater fish have been deliberately introduced to Aotearoa-NZ for aquaculture, sport and tourism (McDowall, 1994; MacNeil *et al.*, 2024). Unfortunately, many of these have had negative impacts on native freshwater fauna through increased predation and competition (Crowl *et al.*, 1992; Jones and Closs, 2017; NIWA, 2020). The deliberate introduction of a catfish to Aotearoa-NZ, the North American brown bullhead *Ameiurus nebulosus* in 1878, has been associated with the decline of several native species (McDowall, 1990, 1994; Barnes and Hicks, 2003). Consequently, this catfish is now

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designated as a 'pest' in five of the country's regions, with its catching, release and propagation all banned (NIWA, 2020).

Kōura can suffer significant predation from opportunistic omnivores such as the brown bullhead catfish and native fish such as eel (the long-finned *Anguilla dieffenbachii* and the short-finned *Anguilla australis*) (McDowall, 1990: Shave *et al.*, 1994; Sagar and Glova, 1998; Clearwater *et al.*, 2014). The catfish was first reported in the North Island's Lake Taupō, Australasia's largest lake, in the 1980s (Barnes and Hicks, 2003) and subsequently northern kōura have become a core component of catfish diets, being found in stomachs of up to 64% of large catfish (>250 mm) caught in rocky areas of the lake (Barnes and Hicks, 2003). Eel predation on crayfish can also be severe, being cited as responsible for the very restricted distribution of crayfish in some river systems in Northern Ireland (Reynolds, 1982) and for the elimination of some crayfish populations in Sweden (Svardson, 1972). In Aotearoa-NZ, eel predation may also have a significant impact on northern kōura populations and can act in conjunction with catfish predation in some systems (Clearwater *et al.*, 2014). For instance, a study investigating fish predation on northern kōura in North Island hydro lakes, indicated that when eels were absent, these kōura populations could sustain themselves in the presence of low densities of catfish, when catfish density is low, or higher combined densities of eels and catfish, kōura population sizes consistently declined (Clearwater *et al.*, 2014).

Along with direct predation impacts, there are non-consumptive effects (NCEs) of fish predator presence that can have energetic costs for crayfish prey, such as predator presence affecting the time individual crayfish spend foraging and using refuge habitats (Adams, 2007; Stewart and Tabak, 2011; Wood and Moore, 2020). Such changes in crayfish behaviour are not only the result of visual and physical cues, but also chemical cues (kairomones) emitted by fish predators (Wood and Moore, 2020). These cues allow crayfish to evaluate predator threats to a high level of precision in the freshwater environment and fine tune their responses accordingly (Wood and Moore, 2020; Musil *et al.*, 2023). For instance, crayfish can even use fish predator odours to assess the relative size ratios of fish predators to gauge the level of threat an unseen fish predator poses (Wood and Moore, 2020).

Currently, many studies of Aotearoa-NZ freshwater food webs when considering the 'role' of kōura in relation to fish, focus almost exclusively on kōura as a prey item, with little or no regard for the potential significance of kōura as an active scavenger of the same fish predators (Brown, 2009). Our study aimed to investigate aspects of scavenging behaviour of kōura on the carrion of native and non-native predators, to highlight how significant this trophic link could be in many Aotearoa-NZ systems. In addition, while coevolved native species of crayfish and fish can be in general ecological balance, sustained by mutual predation and competition for food and spatial resources (Reynolds, 2011; Thomas and Taylor, 2013), this balance can be significantly disrupted by the arrival of non-native crayfish and/or fish species (Reynolds, 2011). Crayfish can detect the difference between fish carrion odour and live fish predator odours (Willman *et al.*, 1994). However, it has also been shown that kōura seem either unable or only to a limited extent, detect and respond to chemical cues emitted by live non-native fish such as brown trout (*Salmo trutta*), whereas they can readily detect and respond to live native fish, such as the long-fin eel (Shave *et al.*, 1994). Therefore, we also aimed to ascertain if there were significant differences regarding kōura scavenging behaviour towards carrion of long-finned eel, a predator sharing a long coevolutionary history with kōura (McDowall, 1964; Brown, 2009) and brown bullhead catfish, a novel predator present for less than one hundred and fifty years and which is currently expanding its range. In addition to kōura orientation and movement in the presence of carrion of coevolved and non-native fish predators, we also investigated kōura feeding rates and calorific assimilation rates of different fish tissue, to ascertain if these differed for co-evolved and non-native fish. In summary, we might expect koura to display significantly different scavenging behaviour with regard to the carcasses of non-native fish carcasses relative to co-evolved native fish carcasses, if kōura readily distinguished between fish predator carcasses based on familiarity or even remnant perceived threat, as opposed to functioning as a purely opportunistic scavenger, simply perceiving all dead fish as the same basic food resource, regardless of origin. Crayfish, including kōura, exhibit their highest activity in darkness (McMahon *et al.*, 2005; Parkyn and Kusabs, 2014) and therefore we tested if koura scavenging behaviours were significantly different in the light and dark.

2 Materials and methods

2.1 Study species

Male and female northern kōura *Paranephrops planifrons* were collected from Brook Stream in The Brook Waimarama Sanctuary, Nelson, New Zealand (lat. 41°31' S, long.173° 29' E) in September 2023. Eels are present in Brook Stream but there have never been records of catfish in this region of Aotearoa-NZ. Unfortunately, kōura were not available from cultured indoor or closed aquaculture systems, as these currently do not exist. It is acknowledged this would have been the ideal scenario, as this would have assured the use of kōura naïve to the chemical cues from any fish predators.

Adult kōura (31.8 g ± 5.1 g wet weight; mean ± SE) were housed in holding tanks (20 animals per 100 L flow through container) in the Cawthron Aquaculture Park aquarium facilities, with broken terracotta pots and 15 cm lengths of drainage pipe providing refuges. While these adults were small, they were all sexually mature and this size range was typical of Brook Stream at the time of sampling. The male: female ratio of the collected adult animals was approximately 70 : 30. Males have a pair of gonads that protrude from the base of the fourth pair of legs, while females have holes at the base of the second pair of legs, their abdomens are flatter and abdominal side plates are larger than males (Chapman et al., 2011). Kōura were acclimatised for two weeks, tank water temperatures were kept near 16.0 °C, on a 11:13 light/dark cycle, a regime appropriate for the time of year. Tank water was an oxygenated 50:50 mix of stream water and dechlorinated tap water. Kōura were fed daily *ad libitum* on commercial farmed salmon fish food pellets (70% vegetable material and 30% marine fish meal/oil). The animals used for experiments were then held separately in smaller 9 L tanks (15 cm × 20 cm × 15 cm) to avoid cannibalism and aggressive interactions during a starvation period of five days, which was undertaken to ensure a standardised motivation to orientate

towards chemical cues / food sources (Epley et al., 2015). All kōura selected for experiments had intact antennae, chelae and legs. No animal was used in more than one choice experiment. For all experiments, recently moulted animals or females in 'berry' (gravid with eggs) were excluded, as these can show altered behaviours (Stewart and Tabak, 2011).

Because kōura are taonga, special permission had to be sought from the relevant iwi (Māori tribal unit), as well as the relevant government regulator, to source a limited number of animals from the Brook Stream. The total number was restricted, as because of biosecurity concerns and Aotearoa-NZ legislative requirements, even taonga animals could not be returned to the Brook Stream after cessation of experiments and had to be euthanized. This was despite experiments being non-invasive and there was no mortality of kōura in any experiment. We thus adapted experiments to make the best use of the limited number of collected animals, while still being scientifically valid.

2.2 Carcasses / Food sources

Brown bullhead catfish were obtained from Lake Rotorua (North Island), long-finned eel from Lake Pearson (Moana Rua, South Island) and Alaskan Pollock (*Theragra chalcogramma*), a marine whitefish, sourced from a local supermarket. The latter species was chosen as a fish species 'control', being an example of a fish that kōura could never encounter in natural systems. All fish carcasses were frozen prior to experiments and defrosted for two days immediately before experiments commenced. Pilot studies showed crayfish would readily consume defrosted fish tissue and other studies have shown defrosted prey remain attractive to crayfish (Gherardi and Barbaresi, 2007). Pilot trials also revealed that even small cubes (2 cm × 2 cm × 2 cm) of fish carcass rapidly disintegrated and plumes of discolouration from cubes of different fish species quickly merged in the confines of a small tank. Therefore, it was decided to adopt the method of Epley *et al.* (2015), for experiments 1 and 2 (see following sections), where fish gelatin blocks were used. These can effectively simulate the stimulus of slowly diffusing carrion to crayfish in freshwater environments and caused far less discolouration and merging of plumes during the experiments. For each species; catfish, eel and pollock, fish gelatin cubes were made by first dissecting each fish so that always longitudinal slices of fillet were cut to standardize the same body part for each species used. Then 60 g of this tissue was homogenized in a blender in 150 ml of cold water. The appropriate amount of unflavoured gelatin (Queen gelatine powder unflavoured) was added to 500 ml of boiling water as per manufacturer's instructions and this was then mixed with the fish homogenate. This mixture was then refrigerated at 3 °C, cut into cubes (2 cm × 2 cm × 2 cm) when set and these cubes refrigerated until time of use. Cubes of each species were wrapped in tinfoil and stored separately from one another in sealed containers, to avoid the potential for cross-contamination.

2.3 Experiment 1–Time to leave refuge habitat in presence of different fish carcasses

In holding tanks, plastic tubes (15 cm long, 6 cm hole diameter, sealed at one end) were observed to be commonly used by kōura as refuge habitat for hours at a time, with individual kōura commonly retreating into tubes if disturbed (see Stewart and Tabak, 2011 for similar findings with *P. zealandicus*). To investigate if fish carcasses from native and introduced species provoked different emergence times for kōura to leave refuge habitats, an individual kōura was placed in a tube and the tube placed in an experimental tank (15 cm × 20 cm × 15 cm) that contained two gelatin fish cubes of one fish species. These cubes were positioned directly in front of the tube entrance, at a distance of 10 cm. The time taken for the kōura to emerge from the tube was then recorded. This set-up was repeated for kōura exposure to either catfish, eel, pollock cubes or gelatin only (no fish cubes) ($n = 5$ treatment replicates) during the light cycle. The set-up was repeated using different kōura in the dark cycle. We did not reuse individuals within a light or dark cycle for different treatments. Red light was used to observe kōura emergence (via video-camera viewer) during the dark cycle, as red light does not influence crustacean behaviour during night-time observations (Weiss *et al.*, 2006) and has been successfully used in recording the anti-predator behaviour of *P. zealandicus* (Shave *et al.*, 1994). Tanks were observed every minute up to 60 min and then once every 30 min, until the final kōura had emerged. We did not distinguish between male and female crayfish in setting up our experiment due to our restricted pool of available test animals and previous studies showing no difference between sexes in behavioural and feeding preference experiments (Shave et al., 1994; Gherardi and Barbaresi, 2007; Martin, 2014).

2.4 Experiment 2–Fish carcass orientation and feeding choice

An individual kōura was placed in a 7.5 cm long plastic tube with a 4 cm hole diameter, sealed at one end. The tube was placed upright in the centre of the experimental tank with the sealed end on the bottom. This tube was placed to position individual kōura in the middle of the tank, while easily allowing kōura to exit in any direction at will. Each corner of the tank held two gelatin fish cubes (gelatin cubes of catfish or eel or pollock or gelatin with no fish), positioned within a 3 cm × 3 cm marked square. This set-up was carried out during the light and dark cycles ($n = 17$ and $n = 14$ respectively) and the location of each individual kōura recorded as either catfish, eel, pollock or gelatin only areas or 'other' (kōura either recorded still within the plastic tube or located in a non-marked area of tank containing no cubes). Observations on kōura location were made 10 min, 30 min, 60 min, 90 min, and 120 min after kōura were placed in the tank. Pilot studies showed that once kōura had left the tube, it took a minimum of 10 min for an individual to manoeuvre itself into one of the four cube zones in the light cycle. All kōura were observed for the 120 min of the experiment.

2.5 Experiment 3 – Feeding rates on different fish carcasses

Finally, animals from experiments 1 and 2 were returned to separate holding tanks when each experiment was terminated and were maintained on commercial farmed salmon fish food pellets for a minimum of five days. These animals were then combined in the same holding tanks, fed pellets for a further five days and then starved for five days. From this pool of animals, fifteen male and fifteen female kōura were taken at

random and used in feeding rate experiments. It is acknowledged that this means just by chance, at least some animals may have been exposed to the same fish species twice (from previously participating in either experiment 1 or 2). However, it is assumed the randomized nature of the kōura selection, the immediately preceding ten days of subsistence on fish food pellets followed by starvation, coupled with the level of experiment replication, would mitigate against the effects of any individual kōura being familiar with the particular fish species being presented to it in a new 'no choice' feeding experiment.

One male kōura (mean wet-weight 38.2 g, range 29.2–41.1 g) and one female koura (mean wet-weight 28.2 g, range 24.5–35.6 g) were added to an experimental tank containing either 20 g wet-weight of catfish, eel or pollock ($n = 5$ treatment replicates). This was well in excess of what other freshwater crayfish species in feeding experiments are capable of consuming, where 4% of bodyweight of food per individual crayfish per day was considered feeding to excess (Nightingale *et al.*, 2021). Cubes of raw fish fillet rather than fish gelatin cubes were deployed, so as to ascertain the exact amount of only fish tissue consumed. Kōura were left to feed for 72 h, encompassing three light and dark cycles. After 72 h the wet-weights of the kōura were recorded, along with the wet-weight of remaining fish tissue. For the latter, this included all fish tissue fragmented but not consumed, which was recovered after filtering experimental tank water. It is acknowledged that this filtrate may have included minute amounts of crayfish faeces from the two individual kōura but it is assumed faeces production during this limited 72 h time-span would be both very small and not differ significantly dependent on the different fish species consumed. It was also assumed that the levels of kōura coprophagy was the same for all replicates.

Although water clarity had slightly diminished by day 3 partially due to leaching, as well as kōura feeding activity / fragmentation, separate controls were undertaken to adjust for leaching losses from fish tissue wet weights ($n = 5$ per fish species). Feeding rates (mg wet weight fish eaten per g wet weight kōura^{-1} 24 h^{-1}) on different fish species were compared (Ruokonen and Karjalainen, 2022). A Parr semi-micro oxygen bomb calorimeter was also used to obtain the calorific value of tissue samples from each fish species ($n = 3$ per fish species) and allow calculation of the kōura calorific assimilation rates for the three fish species (kcal wet weight fish eaten, g wet weight kōura^{-1} 24 h^{-1} in each tank). During the 72 h of the experiment, there was no mortality of kōura during the feeding experiment, kōura were observed actively feeding and no aggressive interactions were observed between males and females.

2.6 Statistical analyses

For experiment 1, we used Cox proportion hazard regressions to analyse the data, via the 'glht' R package, with post-hoc pairwise tests carried out on cube type via simultaneous tests for general linear hypotheses. For experiments 2 and 3, data were analysed using IBM SPSS Statistics (version 26). For experiment 2, we used Friedman's test on kōura distribution specifically with regard to cube orientation (percentage of those kōura which had left tubes and then actively moved into cube zones over the five time intervals of the experiment). Multiple group (pairwise) comparisons were made using the Nemenyi test showed in terms of cube type. For experiment 3, one-way ANOVAs on log transformed data were used to test for significant differences in kōura consumption rate of the three fish species and for kōura calorific assimilation rate between the three fish species. Tukey's HSD test was used for multiple group (pairwise) comparisons.

3 Results

3.1 Experiment 1–Time to leave refuge habitat in presence of different fish carcasses

The time taken for kōura to leave refugia when different fish carcasses are present, are showed as survival curves, which indicate the probability of an event (emergence) occurring through time. The model with light treatment and cube type fit the data significantly better than the null model, (Likelihood Ratio test, χ^2 (2) = 38.01, $p < 0.01$), driven by the slower emergence time in the light cycle as opposed to dark cycle, and for the gelatin cube compared to the three fish species cubes (Fig. 1). In the presence of fish carrion in the light-cycle the fastest emergence time was 6 min and the longest was 45 min, with a mean (±SE) emergence time of 17.3 (±1.3) min. In contrast, in the presence of carrion in the dark cycle, kōura started to emerge after only 3 min and all kōura had emerged after 22 min in the dark cycle, with a mean (±SE) emergence time of 10.2 (±0.6) min.

The mean (±SE) emergence times in the light cycle in the presence of eel and catfish were similar (20.4 (±3.3) min for eels compared to 20.8 (±4.0) min for catfish). In contrast, the emergence time in the presence of pollock was almost half these values (12.07 (±2.0) min). The equivalent times in the dark cycle for eel and catfish were both much lower than in the light cycle, these being 10.4 (±1.9) min and 8.4 (±1.5) min respectively. The emergence time for pollock of 11.9 (±2.0) min in the dark cycle was higher than that of eel or catfish in the dark and was similar to the emergence time for pollock in the light. Post-hoc pairwise tests showed no significant difference in emergence times between any of the three species ($Z = 0.328$, $P = 0.986$; $Z = 0.579$, $P = 0.932$; and $Z = 0.247$, $P = 0.994$ for catfish v. eel, catfish v. pollock and pollock v. catfish respectively, light and dark cycle combined).

The earliest emergence time for an individual in the light cycle when gelatin only cubes were provided, occurred between 90 and 120 min, with the remaining four individual kōura taking between 240 min and 360 min to emerge. The earliest emergence time in the dark cycle was 52 min, with the remaining kōura taking between 90 and 120 min to emerge. Post-hoc pairwise tests showed significant differences in emergence times between all three fish species and gelatin ($Z = 3.562$, $P < 0.01$; $Z = 3.451$, $P = 0.01$; and $Z = 3.603$, $P = 0.0.001$ for catfish, eel and pollock versus gelatin respectively).

3.2 Experiment 2–Fish carcass orientation and feeding choice

After 10 min in the light cycle, eleven out of seventeen kōura still remained within the upturned tubes and it took longer than 90 min for all kōura to leave the tubes. There was a significant difference in the distribution of the kōura between

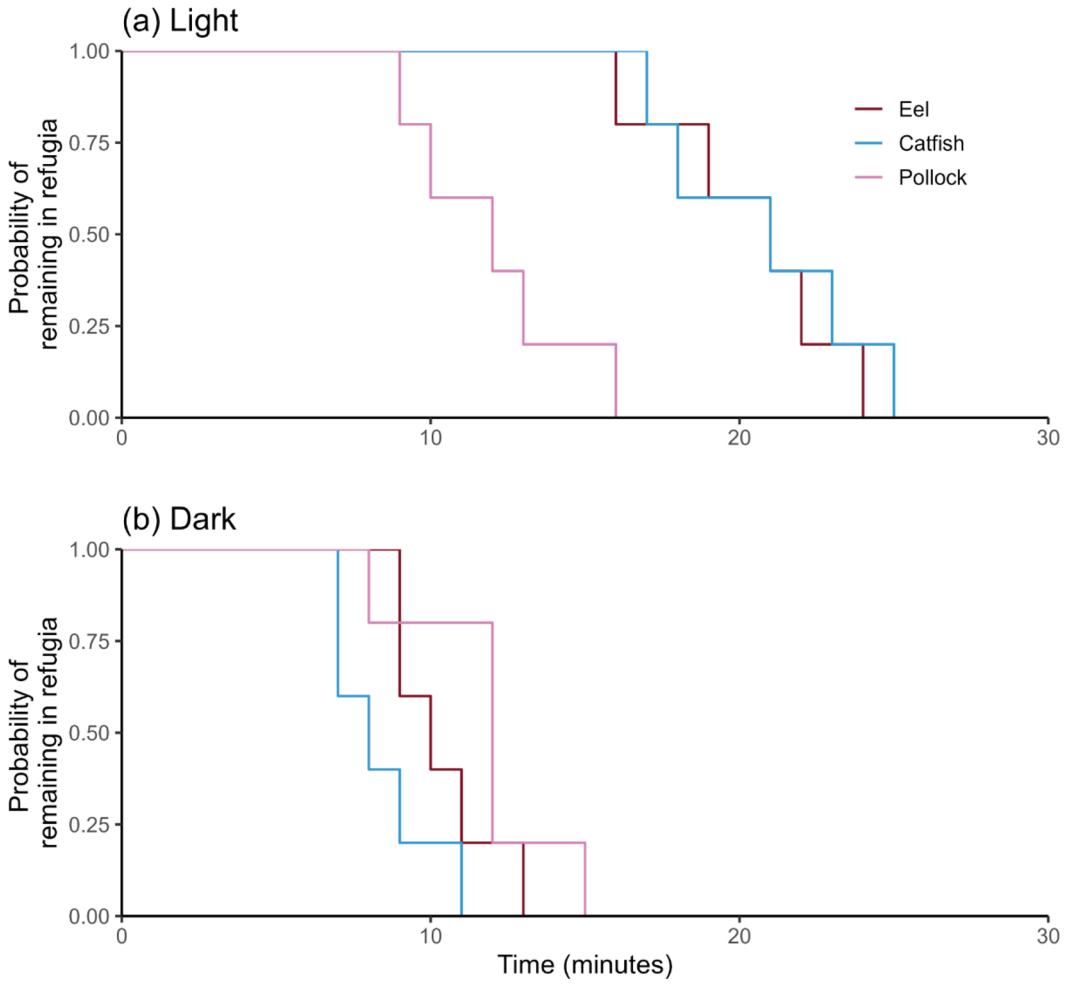

Fig. 1. Time to event curves showing the time for kōura to emerge from refugia based on the presence of different fish species cubes.

different cube types in the light cycle ($\chi^2 = 9.638$, df = 3, $p < 0.05$; Fig. 2a). Multiple group (pairwise) comparisons using the Nemenyi test showed in terms of fish species, there were no significant differences between the three fish species cube types and kōura occurrence ($p = 0.995$ and $Q = 0.346$ for catfish v. eel, $p = 0.999$ and $Q = 0.173$ for both catfish v. pollock and eel v. pollock). Kōura also located themselves more often at catfish cubes rather than gelatin only cubes ($Q = 3.638$, $p < 0.05$). There were also weak trends for kōura to be located more frequently at eel and pollock cubes rather than gelatin only cubes ($Q = 3.464$, $p = 0.068$ and $Q = 3.291$, $p = 0.092$ for eel and pollock respectively). There was only one recorded occurrence of kōura at the gelatin only cube during the entire 120 min timespan of the experiment. There appeared to be strong fish cube fidelity with 12 out of 17 individuals staying at their initial fish species choice during all subsequent time intervals until the end of the experiment. This finding should, however, be treated with caution as this was based on intermittent observations so it was possible that kōura could have moved out of a zone and then back within the space of a time interval.

In the dark cycle, there was also a significant difference in the distribution of the kōura between cube types ($\chi^2 = 13.653$, df = 3, $p < 0.01$; Fig. 2b). In contrast to the light cycle, in the dark cycle all kōura had left tubes and distributed themselves amongst cubes within the first 10 min of the experiment. There were no significant differences between the three fish species cubes and kōura occurrence ($p = 0.827$ and $Q = 1.212$ for catfish v. eel, $p = 0.159$ and $Q = 2.945$ for catfish v. pollock and $p = 0.611$ and $Q = 1.732$ for eel v. pollock). As in the light cycle, kōura occurred more frequently at catfish cubes than gelatin only cubes ($Q = 4.850$, $p < 0.01$). Kōura also occurred significantly more frequently at eel cubes than gelatin only cubes ($Q = 3.640$, $p < 0.05$). There was no trend for kōura to occur more frequently at pollock cubes than gelatin only cubes ($Q = 1.905$, $p = 0.533$). There were only two recorded occurrences of kōura at the gelatin only tubes during the entire experiment. There was strong fish cube fidelity, with 9 out of 14 individuals staying at their initial fish species choice during all subsequent time intervals until the end of the experiment.

At the end of both the light and dark cycle experiments, all fish cubes showed visual evidence of feeding or attempted feeding with scrape and puncture marks, but cubes generally appeared more shredded and scraped in the dark cycle. All gelatin only cubes appeared intact.

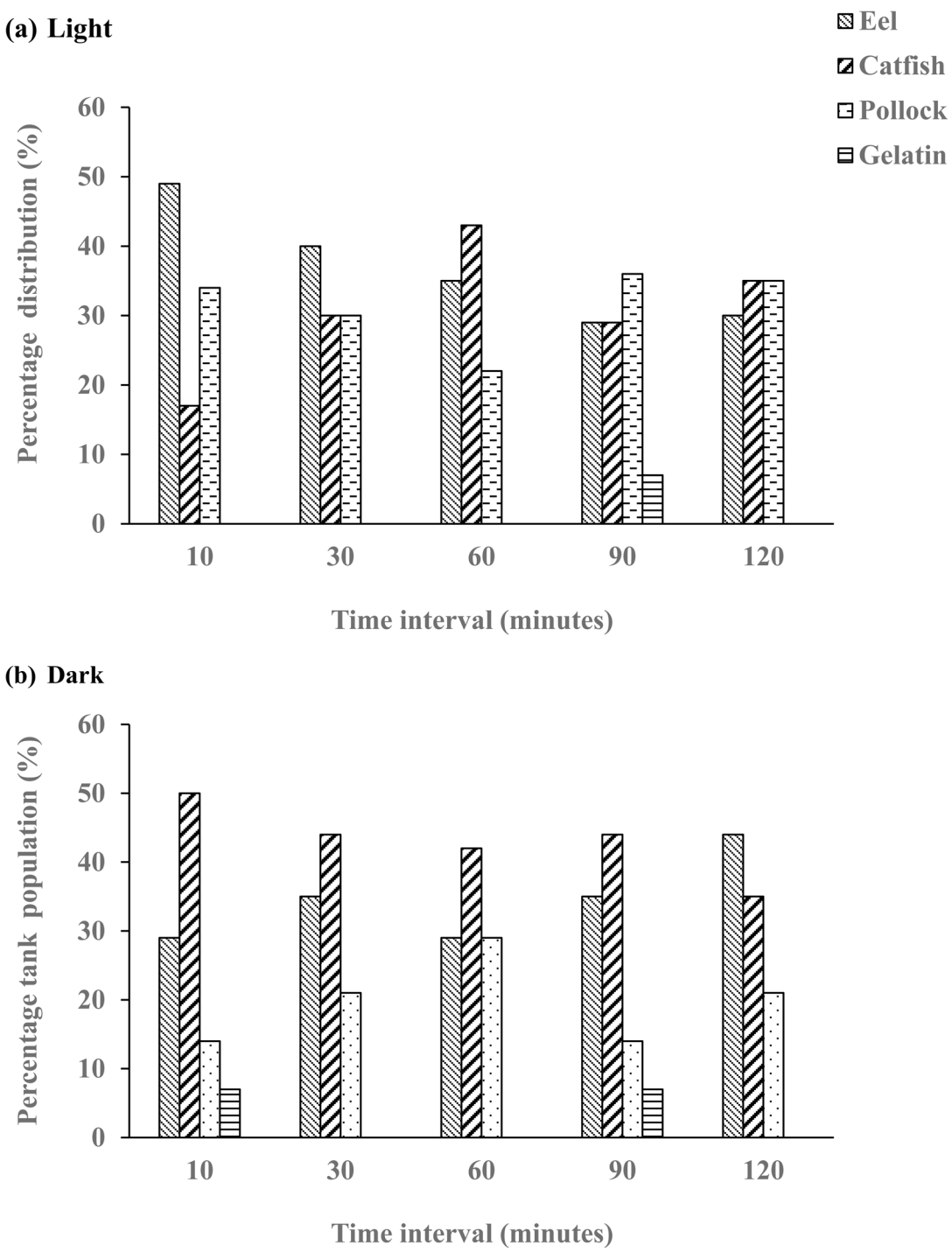

Fig. 2. Percentage distribution (%) of kōura amongst fish species or gelatin only cubes at selected time intervals over 120 min. Note untransformed raw data is presented for clarity and represents the collective distribution of all kōura who had left tubes and actively located themselves in cube areas.

3.3 Experiment 3- Feeding rates on different fish carcasses

There was no significant difference in kōura feeding rate of the three fish species (mg wet weight, g wet weight^{-1} 24 h^{-1}) ($F_{2,12}$ = 1.230, NS, p = 0.327) (Fig. 3a). Ash-free dry weight (AFDW) calorific values were obtained for the three fish species, these being 96 kcal/g for catfish, 179 kcal/g for eel and 71 kcal/g for pollock. There was a significant difference in kōura calorific assimilation rate between the three species (kcal wet weight, g wet weight^{-1} 24 h^{-1}) ($F_{2,12}$ = 5.743, $p < 0.05$) (Fig. 3b). The calorific assimilation rate was higher for eel than

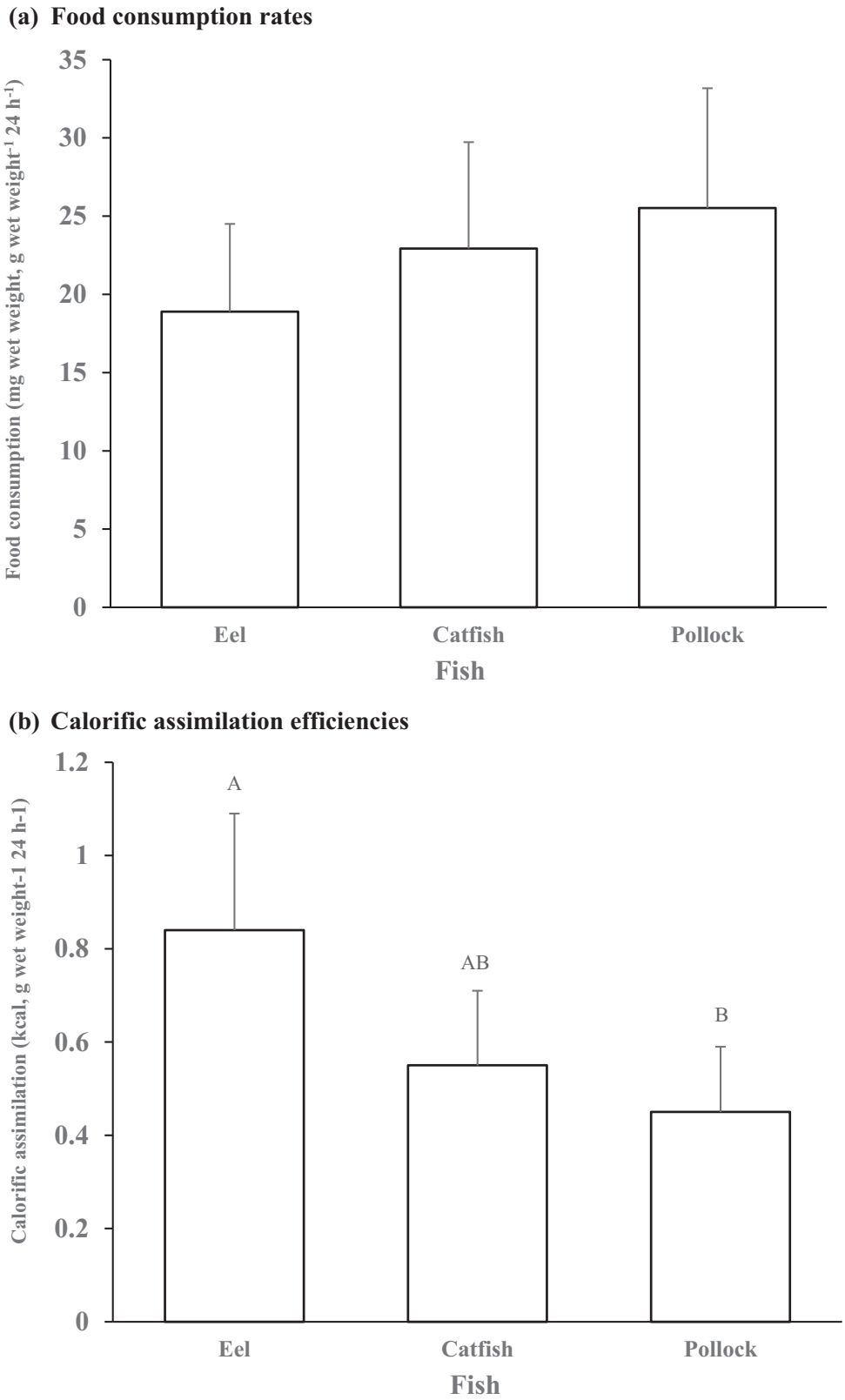

Fig. 3. Mean (± standard error) (a) food consumption rates (mg wet weight fish eaten, g wet weight kōura^{-1} 24 h^{-1}) and (b) calorific assimilation efficiencies (kcal of fish eaten, g wet weight kōura^{-1} 24 h^{-1}) of eel, catfish or pollack by kōura. For (b), significant differences between treatments are emphasised with letters.

pollock (Tukey's HSD test $t = 4.603$, $p < 0.05$) but there was no significant difference between catfish and pollock ($t = 1.143$, NS). There was also no significant difference between eel and catfish ($t = 3.460$, NS). Controls confirmed the offered fish tissue types were subject to slight changes in weight due to leaching (Gherardi and Barbaresi, 2007), during the 72 h of the experiment, but that the leaching losses were similar for all fish species and these were adjusted for (mean losses of 1.33%, 1.00% and 1.00% for eel, catfish and pollock controls respectively).

4 Discussion

These experiments emphasise the fact that freshwater crayfish such as kōura, are not just prey for fish, they may also have an important role as scavengers of native and introduced fish carcasses, in freshwater ecosystems in Aotearoa-NZ. This is unsurprising as the importance of freshwater crayfish in the role as active scavengers and energy cycling has been shown in other freshwater systems (Guan and Wiles, 1998; Boros *et al.*, 2020). There also appears little discrimination between the fish species the carcass belongs to as regards the scavenging / feeding behaviour of kōura. Whether the fish species is a native – coevolved with kōura, an invader – novel to kōura or even a marine species that can never come into contact with kōura under natural conditions, our study shows kōura will orientate towards any dead fish tissue and readily consume it. The role of crayfish as opportunistic scavengers of both native and non-native fish carcasses was also evident in a field experiment by Unger and Hickman (2019). That study showed that North American native crayfish species would actively seek out and scavenge trout carcasses within hours of their deployment in streams, regardless of whether they were from native (brook trout *Salvelinus fontinalis*) or non-native species (rainbow trout *Oncorhychus mykiss*).

Martin (2014) showed that farmed red swamp crayfish (*Procambarus clarkii*) who were predator naïve, showed reduced antipredator behaviour and survivorship to a predator (the largemouth bass *Micropterus salmoides*) they had never encountered before. This manifested itself in a lack of crayfish responses to chemical cues from bass in terms of appropriate use of refuges, as well as crayfish movement (Martin, 2014). Shave *et al.* (1994) showed that southern kōura (*P. zealandicus*) were able to use chemical cues from mucus to detect native fish such as eels but not introduced non-native brown trout (*Salmo trutta*). This may mean kōura species are at greater risk from living introduced predators than living native predators, because of this inability to detect non-contact chemical cues. Our study indicates that while that may be true for living predators, when a fish is dead, kōura may not perceive it as a predatory risk but rather may perceive it as a potential food source and something which can trigger leaving of a refuge habitat. However, it is possible that whatever chemical cues exist that elicit a kōura 'fear' non-consumptive effect response in the presence of living predators (Wood and Moore, 2020) are either not given off by dead fish or are overwhelmed by other chemical cues that elicit a feeding response. This latter point may be evident in the study by Unger and Hickman (2019), which showed that even in high density trout streams, crayfish would still readily scavenge trout carcasses, despite the simultaneous constant predation threat from live trout.

Although emergence from refuges and orientation towards fish cubes occurred at a slower pace in the light than dark cycle, it still occurred for all individuals, even if it took several hours under artificial light. In general, freshwater crayfish are known to be negatively phototactic and most active during the dark (Unger and Hickman, 2019; Gherardi *et al.*, 2000; Gherardi, 2002), although kōura have been observed active in the daylight in small streams lacking fish predators (Holmes, pers. obs.). Thomas *et al.* (2016) and Jackson and Moore (2019) found that freshwater crayfish were negatively impacted by artificial light pollution, becoming less active. Species such as signal crayfish (*Pacifastacus leniusculus*) being mainly nocturnal, normally showed peak activity and interaction levels during darkness, taking refuge in daylight, but simulated light pollution altered this activity pattern, causing crayfish to spend far longer in refuges. Our results confirm that kōura are generally nocturnal but the presence of potential food sources in the immediate vicinity and lack of perceived predation threat (*i.e* no live fish predators) can trigger active scavenging behaviour even in light conditions that could it vulnerable to any visual predators, if repeated in natural systems.

In terms of kōura feeding, the calorific assimilation efficiency for eel was the highest of the three species and significantly higher than pollock. This is unsurprising as eels tend to be rich in fat, protein, minerals and vitamins compared to many other fish species (Seo *et al.*, 2013). Scavenging of eel carcasses by kōura is seldom if ever reported in Aotearoa-NZ but our study indicates this is probably constantly occurring within natural systems and eels (themselves a taonga species) would represent a nutritious food resource for kōura. In terms of overall kōura feeding rate, there was no significant difference between the three fish species, indicating that kōura are probably already actively scavenging the carcasses of catfish which have died naturally, within invaded river and lake ecosystems in Aotearoa-NZ, even if this particular trophic interaction is currently underappreciated or not recognised at all. For instance, in the seminal paper on the Lake Taupō invasion by catfish, there is a detailed account of the feeding habitats of the increasingly abundant catfish population and its impact on resident kōura but there is no consideration given to the fact that the kōura themselves may be scavenging carcasses of any dead catfish (Barnes and Hicks, 2003).

Our study is the first showing scavenging behaviour of kōura on catfish and showing no discrimination in crayfish feeding behaviour on native and introduced fish predators in Aotearoa-NZ. Showing kōura will readily feed on dead catfish has important practical implications. Catfish are having a serious impact on kōura populations in Aotearoa-NZ (Barnes and Hicks, 2003) and this has led to various catfish eradication programmes involving regional councils, an iwi (Māori tribal unit) governed Lakes Trust (the Te Awara Lakes Trust, which is responsible for protecting and restoring 14 lakes in the Waikato and Bay of Plenty regions) and iwi led citizen science groups such as the 'Catfish Killas'. The overall eradication programme in the Te Awara Lakes area has culled 180,000 catfish in the eight years since it was established in 2016 (https://tearawa.io/programmes/). There have been limited attempts to commercialise catfish takes in Aotearoa-NZ by selling these to the Asian market (where these might be considered a delicacy, as in the southern states of USA, where they originated (Brown, 2001)), and to repurpose large culls as

fertiliser (Scott-Simmonds 2021). However, such attempts have been unsuccessful and dead catfish as a basic resource have not yet been successfully commercially exploited. Currently, culled catfish are simply buried on a routine basis. Kōura represent one of the earliest types of freshwater fishery in Aotearoa-NZ and as taonga species still support important Māori customary fisheries in several North Island lakes (McDowall, 2011). Commercial freshwater crayfish farming is a global activity (Holdich, 1993) but in Aotearoa-NZ, kōura aquaculture remains in early developmental stages, despite farmed kōura potentially representing a high-value novel product for an evolving primary sector in Aotearoa NZ (Hollows, 2016, 2020). Our study indicates the carcasses of culled catfish, could represent a high protein, nutrient-rich resource for kōura, for instance as fishmeal, in any future aquaculture setting. Other non-native fish such as brown trout (*Salmo trutta*) make effective bait for catching kōura (Kusabs, pers. obs.), and it is likely that catfish would also be equally as effective as bait. We suggest, with suitable precautions, some of the thousands of culled catfish from ongoing control and eradication programmes, could therefore be repurposed in a number of ways, as a high quality and plentiful food source for kōura. This could help offset the negative impacts that this catfish invader is currently having on native kōura populations.

5 Conclusions

Our experiments showed kōura readily leave refugia to scavenge on the carcasses of dead predatory fish, with little discrimination between native eel, invasive catfish or marine (control) fish. There was no difference in kōura consumption rate of different fish and no difference in the calorific assimilation rate between eel and catfish. Our results suggest kōura are scavenging the carcasses of catfish within invaded systems, even if this trophic interaction still remains unacknowledged in freshwater food web studies. Catfish eradication programmes in Aotearoa-New Zealand result in culls of thousands of catfish annually and we suggest these carcasses could be repurposed as a food resource for kōura, in a future conservation-based aquaculture setting.

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Conflicts of interest

The authors declare no competing interests.

Author contribution statement

Conceptualisation: CM. Formal analysis: CM, FL. Fieldwork: CM, FL, RH, IK. Investigation: CM. Writing original draft: CM. All authors contributed and approved the final manuscript.

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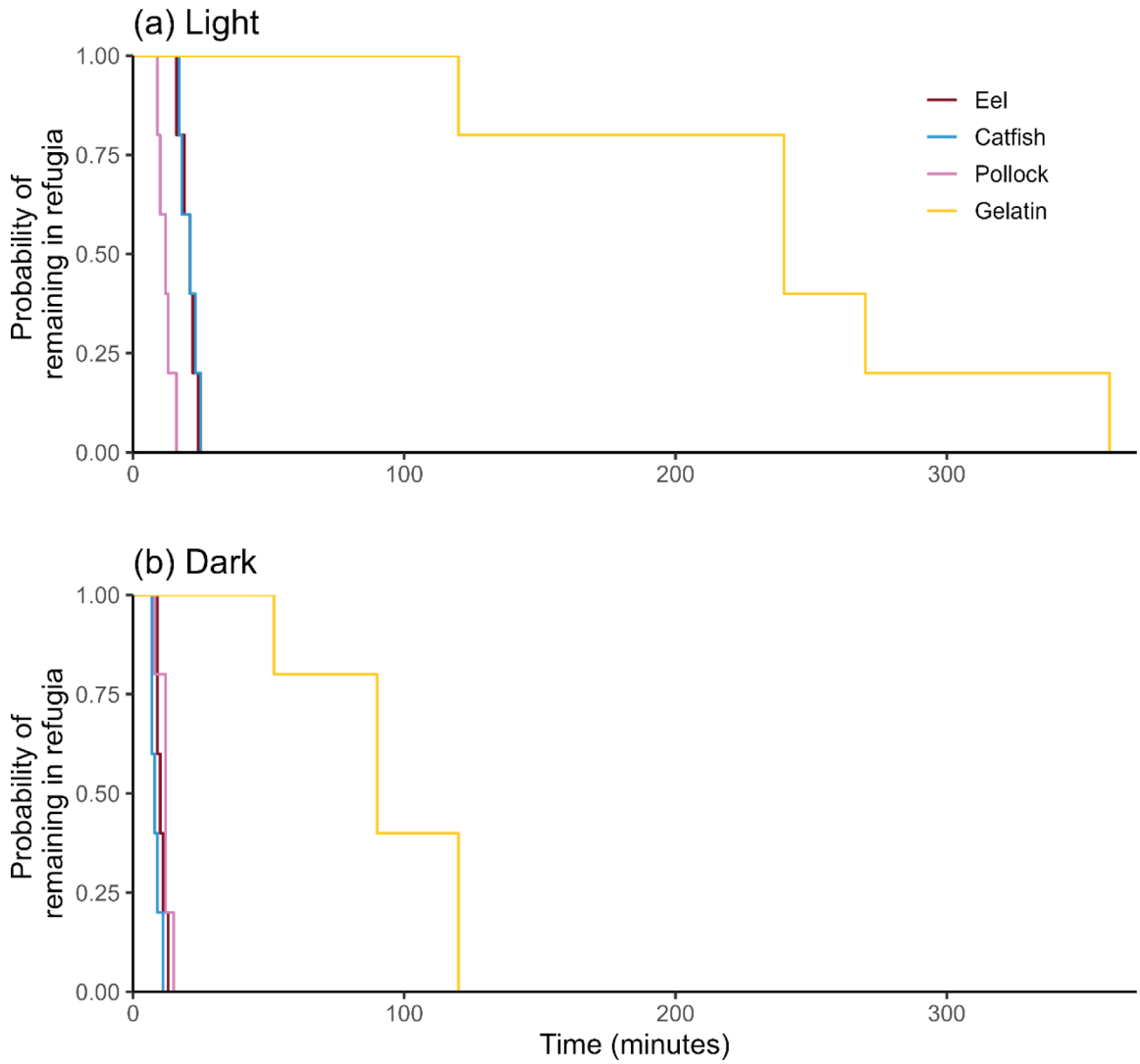
Appendix A

Time to event curves showing the time for kōura to emerge from refugia based on the presence of different food items (three fish species cubes and gelatin only cubes).